

Experiment NO : 04

Experiment Name : Study of the DC biasing circuit of BJT

Objectives:

- To implement BJT biasing circuits
- To set the operating point of a BJT at the mid point of the characteristics curve

Theory:

A Bipolar Junction Transistor (BJT) is a semiconductor device with three terminals that can be configured for signal amplification or digital switching. For a transistor to function reliably, it must be biased so that its operating point falls within a specific region of its characteristics curve—cutoff, active and saturation. The active region is essential for amplification whereas the cutoff and saturation are used when the device needs to behave as a switch.

Types of BJT Biasing Circuits:

- Fixed Bias
- Collector-to-Base Bias
- Voltage-Divider Bias

Among the above mentioned biasing circuits, voltage-divider circuit provides better stability of the operating point. Hence it is preferred over other biasing techniques.

From the circuit diagram, the following equations are obtained,

$$R_{Th} = R_1 \parallel R_2$$

$$V_{Th} = V_{CC} \times \frac{R_2}{R_1 + R_2}$$

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Required Apparatus:

- BJT (BD135, 1 piece)
- Resistors ($1\text{ k}\Omega$, $2.2\text{ k}\Omega$, $4.7\text{ k}\Omega$: 1 piece each)
- Potentiometer ($100\text{ k}\Omega$: 1 piece)
- DC power supply (0-15 V)
- Project Board
- Multimeter
- Connecting wires

Circuit Diagram:

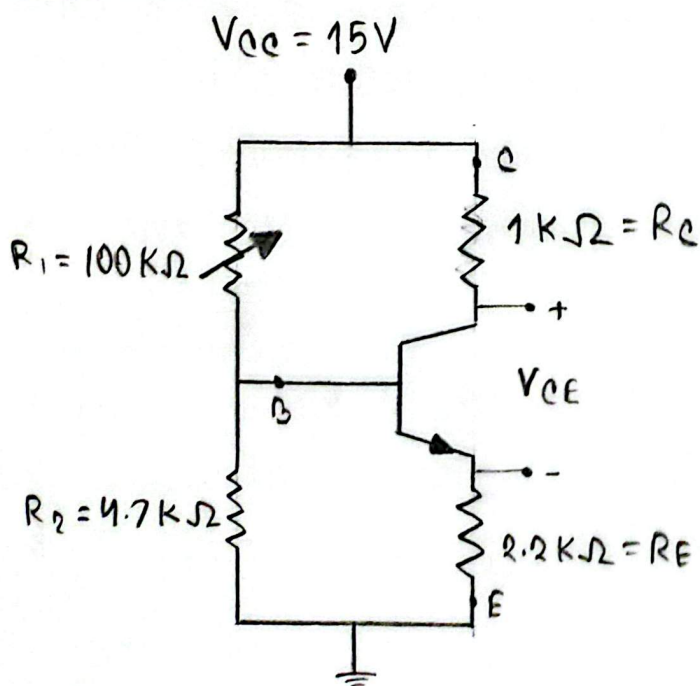


Figure: Circuit Diagram for BJT voltage divider biasing circuit

Procedure:

- (i) Connected the circuit according to the circuit diagram
- (ii) Gradually varied the potentiometer so that V_{CE} becomes equal to the $V_{CC}/2$
- (iii) Measured V_{RC} , V_{BE} , I_B , I_C and β
- (iv) Compared the measured values with the theoretical values

Data Table:

	V_{CE} (V)	V_{RC} (V)	V_{BE} (V)	I_B (mA)	I_C (mA)
Theoretical	7.5	6.15	0.7	17.803	6.15
Experimental	7.45	6.32	0.66	16.5	5.7

Here, $\beta = I_C / I_B = 345.45$

Discussion and Conclusion:

This experiment confirmed that the voltage-divider bias sets a stable midpoint operating point for the BJT, with measured values closely matching theory. Minor differences arose from component variations, but overall the circuit performed as expected.

Precautions:

- All the connections were properly given
- Transistor terminals were identified properly
- Calibration were performed properly